ALL-IN-ONE

1 MEASUREMENTS

1.1 Errors

- Fractional Error: If $Z = \frac{A^x B^y}{C^z}$, then $\frac{\Delta Z}{Z} = x\frac{\Delta A}{A} + y\frac{\Delta B}{B} + z\frac{\Delta C}{C}$
- Absolute Error: If $Z = A \pm B \implies \Delta Z = \Delta A + \Delta B$

1.2 Micrometer Screw Gauge

$$\text{Least Count} \quad LC = \frac{p}{N}$$
$$\text{Total Reading} \quad R = \text{MSR} + (\text{CSC} \times LC)$$

- p: Pitch (Distance moved on linear scale for 1 full rotation of circular scale)
- N: Total divisions on the circular scale
- CSC: Circular Scale Coincidence

1.3 Spherometer

$$\text{Radius of Curvature} \quad R = \frac{a^2}{6h} + \frac{h}{2}$$

- a: Mean distance between the three pointed legs (m)
- h: Height or depth measured by the central screw (m)

2 MECHANICS

2.1 Vectors

Resultant of two vectors:

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

- R: Resultant vector magnitude
- P, Q: Magnitudes of vectors
- θ: Angle between P and Q

2.2 Linear Motion

$$v = u + at$$
$$s = \left(\frac{u+v}{2}\right)t$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

- s: Displacement (m)
- u, v: Initial / final velocity ($\mathrm{m\,s^{-1}}$)
- a: Constant acceleration ($\mathrm{m\,s^{-2}}$)
- t: Time (s)

2.3 Dynamics & Newton's Laws

$$p = mv$$
$$F = ma$$
$$F = \frac{mv - mu}{t} = \frac{\Delta p}{\Delta t}$$
$$J = F\Delta t = \Delta p \quad \text{(Impulse)}$$

- p: Momentum ($\mathrm{kg\,m\,s^{-1}}$)
- F: Force (N)
- J: Impulse (N s)

2.4 Friction

$$F_{s(\max)} = \mu_s R \quad \text{(Static max)}$$
$$F_k = \mu_k R \quad \text{(Kinetic)}$$

- F: Frictional force (N)
- R: Normal reaction force (N)
- μ_s, μ_k: Friction coefficients

2.5 Work, Energy & Power

$$W = Fx$$
$$E_p = mgh$$
$$F = kx \quad \text{(Hooke's Law)}$$
$$E_{ep} = \frac{1}{2}kx^2$$
$$E_k = \frac{1}{2}mv^2$$
$$P = \frac{W}{t} = Fv$$
$$\text{Eff. } (\%) = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%$$

- W, E: Work and Energy (J)
- x: Distance / extension (m)
- k: Spring constant ($\mathrm{N\,m^{-1}}$)
- P: Power (W)

2.6 Rotational & Circular Motion

$$s = r\theta$$
$$\omega = \frac{\Delta\theta}{\Delta t}$$
$$\alpha = \frac{\omega - \omega_0}{t}$$
$$\omega = \omega_0 + \alpha t$$
$$\theta = \left(\frac{\omega + \omega_0}{2}\right)t$$
$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$
$$\omega^2 = \omega_0^2 + 2\alpha\theta$$
$$v = r\omega$$
$$a_t = r\alpha$$
$$a_r = v\omega = \frac{v^2}{r} = r\omega^2$$

- θ: Angular disp. (rad)
- ω_0, ω: Angular vel. ($\mathrm{rad\,s^{-1}}$)
- α: Angular acc. ($\mathrm{rad\,s^{-2}}$)
- r: Radius (m)
- a_t, a_r: Tangential / Centripetal acc.

2.7 Rigid Body Dynamics

$$I = \sum m_i r_i^2$$
$$L = I\omega$$
$$I_1\omega_1 = I_2\omega_2$$
$$\tau = F \times r_{\perp} = I\alpha$$
$$W_{rot} = \tau\theta$$
$$E_{k(rot)} = \frac{1}{2}I\omega^2$$
$$P_{rot} = \tau\omega$$
$$E_{tot} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

- I: Moment of Inertia ($\mathrm{kg\,m^2}$)
- L: Angular momentum ($\mathrm{kg\,m^2\,s^{-1}}$)
- τ: Torque (N m)

2.8 Fluid Statics & Dynamics

$$\rho = \frac{m}{V}$$
$$P = \frac{F}{A}$$
$$P = P_0 + \rho gh$$
$$\Delta P = \rho gh$$
$$U = V\rho g$$
$$mg = U \quad \text{(Floatation)}$$
$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{Const}$$
$$A_1v_1 = A_2v_2$$

- ρ: Density ($\mathrm{kg\,m^{-3}}$)
- P: Pressure (Pa)
- P_0: Pressure at the liquid surface (Pa)
- ΔP: Pressure increase due to depth (Pa)
- U: Upthrust (N)
- v: Fluid velocity ($\mathrm{m\,s^{-1}}$)

3 WAVES & OSCILLATIONS

3.1 Simple Harmonic Motion (SHM)

$$T = \frac{2\pi}{\omega}$$
$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$
$$a = -\omega^2 x$$
$$v = \omega\sqrt{A^2 - x^2}$$
$$v_{\max} = A\omega \quad \text{(at center)}$$

- T: Period (s), f: Frequency (Hz)
- ω: Angular frequency ($\mathrm{rad\,s^{-1}}$)
- a: Acceleration ($\mathrm{m\,s^{-2}}$)
- x: Displacement from equilibrium (m)
- A: Amplitude (m)

Energy in SHM:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$$
$$E_p = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2$$
$$E_{tot} = E_k + E_p = \frac{1}{2}m\omega^2 A^2 \quad \text{(Const)}$$

Pendulums & Springs:

$$F = -kx \quad \text{(Restoring force)}$$
$$T = 2\pi\sqrt{\frac{L}{g}} \quad \text{(Simple pendulum)}$$
$$T = 2\pi\sqrt{\frac{m}{k}} \quad \text{(Mass-spring system)}$$
$$k_{eq} = k_1 + k_2 \quad \text{(Springs in parallel)}$$
$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} \quad \text{(Springs in series)}$$

- k: Spring constant ($\mathrm{N\,m^{-1}}$)
- L: Length of pendulum (m)
- m: Mass (kg)

3.2 Wave Mechanics

$$v = f\lambda$$
$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \quad \text{(Gas)}$$
$$v = \sqrt{\frac{Y}{\rho}} \quad \text{(Solid)}$$
$$v = \sqrt{\frac{T}{m/L}} = \sqrt{\frac{T}{\mu}} \quad \text{(String)}$$

- v: Wave speed ($\mathrm{m\,s^{-1}}$)
- λ: Wavelength (m)
- γ: Ratio of specific heats (C_p/C_v)
- P: Pressure (Pa), ρ: Density ($\mathrm{kg\,m^{-3}}$)
- R: Universal gas constant
- T: Absolute temp (K) / Tension (N)
- M: Molar mass ($\mathrm{kg\,mol^{-1}}$)
- Y: Young's Modulus (Pa)
- μ: Mass per unit length ($\mathrm{kg\,m^{-1}}$)

3.3 Sound & Doppler Effect

$$I = \frac{E}{At} = \frac{P}{A}$$
$$I = \frac{P}{4\pi r^2} \quad \text{(Point source)}$$
$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$$
$$\Delta\beta = 10\log_{10}\left(\frac{I_2}{I_1}\right)$$
$$f_b = |f_1 - f_2| \quad \text{(Beat frequency)}$$
$$f' = f\left(\frac{v \pm v_o}{v \mp v_s}\right) \quad \text{(Doppler)}$$

- I: Intensity ($\mathrm{W\,m^{-2}}$)
- P: Power of source (W)
- β: Intensity level (dB)
- I_0: Threshold of hearing ($10^{-12}\,\mathrm{W\,m^{-2}}$)
- f': Apparent frequency (Hz)
- v: Speed of sound ($\mathrm{m\,s^{-1}}$)
- v_o, v_s: Velocity of observer / source
  Sign convention: Top signs (+ num, - den) for moving towards. Bottom signs for moving away.

3.4 Stationary Waves & Resonance

$$f_n = \frac{nv}{2L} \quad \text{(String / Open pipe)}$$
$$f_n = \frac{nv}{4L} \quad \text{(Closed pipe, n is odd)}$$
$$L_{eff} = L + e \quad \text{(Closed pipe end corr.)}$$
$$L_{eff} = L + 2e \quad \text{(Open pipe end corr.)}$$

- n: Harmonic number
- e: End correction ($e \approx 0.6r$)

3.5 Geometrical Optics

$$_1n_2 = \frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$
$$n_1\sin i = n_2\sin r \quad \text{(Snell's Law)}$$
$$\sin C = \frac{n_{\text{rare}}}{n_{\text{dense}}} \quad \text{(Critical ang.)}$$
$$\frac{\text{Real Depth}}{\text{App. Depth}} = \frac{n_1}{n_2}$$
$$d = t\left(1 - \frac{1}{n}\right)$$

- d: App. displacement
- n: Refractive index
- C: Critical angle
- t: Thickness of medium (m)

Prisms:

$$A = r_1 + r_2$$
$$d = i_1 + i_2 - A$$
$$d_{\min} = 2i - A \quad \text{(at min deviation)}$$
$$n = \frac{\sin\left(\frac{A + d_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Lenses:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{(Lens equation)}$$
$$P = \frac{1}{f}$$
$$\frac{1}{F_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} \implies P_{eq} = P_1 + P_2$$
$$m = \frac{v}{u} = \frac{h_i}{h_o} \quad \text{(Linear Mag.)}$$

3.6 Optical Instruments

Simple Microscope:

$$M = \frac{D}{f} \quad \text{(Abnormal - Image at } \infty\text{)}$$
$$M = 1 + \frac{D}{f} \quad \text{(Normal - Image at } D\text{)}$$

Compound Microscope:

$$M = m_o \times m_e$$
$$M = \left(\frac{v_o}{u_o}\right)\left(\frac{D}{f_e}\right) \quad \text{(Abnormal)}$$
$$M = \left(\frac{v_o}{u_o}\right)\left(1 + \frac{D}{f_e}\right) \quad \text{(Normal)}$$

Astronomical Telescope:

$$M = \frac{f_o}{f_e} \quad \text{(Normal adj.)}$$
$$M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) \quad \text{(Abnormal adj.)}$$

- M: Angular magnification
- D: Least distance of distinct vision ($\sim$ 25cm)
- f_o, f_e: Focal length of objective / eyepiece

4 THERMAL PHYSICS

4.1 Thermometry

$$\frac{\theta - \theta_0}{\theta_{100} - \theta_0} = \frac{X_\theta - X_0}{X_{100} - X_0}$$
$$T = 273.16\left(\frac{X}{X_{tr}}\right)$$
$$T(\mathrm{K}) = \theta(^\circ\mathrm{C}) + 273.15$$

- θ: Celsius temp, T: Absolute temp (K)
- X: Thermometric property (length, resistance, etc.)
- X_{tr}: Property at triple point of water

4.2 Thermal Expansion

$$L = L_0(1 + \alpha\Delta\theta)$$
$$A = A_0(1 + \beta\Delta\theta)$$
$$V = V_0(1 + \gamma\Delta\theta)$$
$$\beta = 2\alpha \quad \text{and} \quad \gamma = 3\alpha$$
$$\gamma_{real} = \gamma_{app} + \gamma_{vessel}$$
$$\rho = \frac{\rho_0}{1 + \gamma\Delta\theta} \approx \rho_0(1 - \gamma\Delta\theta)$$

- α, β, γ: Linear, areal, and volume expansivities ($\mathrm{K^{-1}}$ or $^\circ\mathrm{C^{-1}}$)
- ρ: Density ($\mathrm{kg\,m^{-3}}$)
- $\Delta\theta$: Change in temperature

4.3 Calorimetry

$$Q = mc\Delta\theta = C\Delta\theta$$
$$Q = mL$$
$$\frac{dQ}{dt} = mc\frac{d\theta}{dt} \quad \text{(Rate of heating)}$$

- Q: Heat energy (J)
- c: Specific heat capacity ($\mathrm{J\,kg^{-1}\,K^{-1}}$)
- C: Heat capacity ($C = mc$) ($\mathrm{J\,K^{-1}}$)
- L: Specific latent heat ($\mathrm{J\,kg^{-1}}$)

4.4 Heat Transfer

Conduction:

$$\frac{dQ}{dt} = kA\frac{(\theta_1 - \theta_2)}{d} = kA\frac{\Delta\theta}{\Delta x}$$

- dQ/dt: Rate of heat flow (W)
- k: Thermal conductivity ($\mathrm{W\,m^{-1}\,K^{-1}}$)
- $\Delta\theta/\Delta x$: Temperature gradient ($\mathrm{K\,m^{-1}}$)

Newton's Law of Cooling

$$\frac{dQ}{dt} = eA(\theta - \theta_{room})$$

- e: Convection coefficient

4.5 Thermodynamics

$$\Delta Q = \Delta U + \Delta W \quad \text{(First Law)}$$
$$\Delta W = P\Delta V \quad \text{(Work done by gas)}$$

- ΔQ: Heat added *to* system
- ΔU: Increase in internal energy
- ΔW: Work done *by* system

4.6 Gas Laws & Kinetic Theory

$$PV = nRT = Nk_BT$$
$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad \text{(Combined Gas Law)}$$
$$P = \frac{1}{3}\frac{Nm}{V}\overline{c^2} = \frac{1}{3}\rho\overline{c^2}$$
$$c_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}}$$
$$E_k = \frac{1}{2}m\overline{c^2} = \frac{3}{2}k_BT$$

- n: Number of moles
- N: Number of molecules
- R: Universal gas constant ($8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$)
- k_B: Boltzmann constant ($1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$)
- c_{rms}: Root mean square speed
- M: Molar mass ($\mathrm{kg\,mol^{-1}}$)
- E_k: Average kinetic energy per molecule

4.7 Hygrometry

$$AH = \frac{m_v}{V} \quad \text{(Absolute Humidity)}$$
$$RH = \frac{m_v}{m_s} \times 100\% = \frac{P_v}{SVP} \times 100\%$$

- AH: Mass of water vapor per unit volume
- RH: Relative Humidity
- m_v, m_s: Actual mass / Saturation mass of vapor
- P_v: Partial pressure of water vapor
- SVP: Saturated vapor pressure at same temp

5 GRAVITATIONAL FIELDS

5.1 Gravitation & Fields

$$\text{Force} \quad F = G\frac{m_1m_2}{r^2}$$
$$\text{Field Intensity} \quad g = \frac{F}{m} = \frac{GM}{r^2}$$
$$\text{Surface Intensity} \quad g_o = \frac{GM}{R^2}$$
$$\text{Potential} \quad V = \frac{W}{m} = -\frac{GM}{r}$$
$$\text{Potential Energy} \quad W = -\frac{GMm}{r}$$

- G: Gravitational const. ($6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$)
- M: Mass of planet/star (kg)
- m: Mass of object (kg)
- R: Radius of planet (m)
- r: Distance from center ($r = R + h$) (m)
- V: Potential ($\mathrm{J\,kg^{-1}}$)

5.2 Orbits & Satellites

$$\text{Escape Velocity} \quad v_e = \sqrt{\frac{2GM}{R}}$$
$$v_e = \sqrt{2g_oR}$$
$$\text{Orbital Velocity} \quad v_o = \sqrt{\frac{GM}{r}}$$
$$\text{Kepler's 3rd Law} \quad T = 2\pi\sqrt{\frac{r^3}{GM}}$$
$$\implies T^2 \propto r^3$$
$$\text{Kinetic Energy} \quad E_k = \frac{GMm}{2r}$$
$$\text{Total Energy]} \quad E_{tot} = U + E_k$$
$$E_{tot} = -\frac{GMm}{2r}$$

- v_e, v_o: Velocities ($\mathrm{m\,s^{-1}}$)
- T: Orbital period (s)
- *Note: Geostationary satellite has $T \approx 24$ hours and orbits over the equator.*

6 STATIC ELECTRIC FIELDS

6.1 Coulomb's Law & Electric Fields

$$F = \frac{1}{4\pi\varepsilon}\frac{q_1q_2}{r^2}$$
$$E = \frac{F}{q} = \frac{1}{4\pi\varepsilon}\frac{Q}{r^2} \quad \text{(Point charge)}$$
$$\varepsilon = \varepsilon_r\varepsilon_0 = k\varepsilon_0$$

- q, Q: Electric charges (C)
- E: Electric field intensity ($\mathrm{N\,C^{-1}}$ or $\mathrm{V\,m^{-1}}$)
- ε: Permittivity of medium ($\mathrm{F\,m^{-1}}$)
- ε_0: Permittivity of free space (8.85×10^{-12})
- ε_r, k: Relative permittivity / Dielectric constant

6.2 Gauss's Law & Charge Distributions

$$\lambda = \frac{Q}{L}, \quad \sigma = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$
$$\Phi = EA$$
$$\Phi = \frac{Q_{enc}}{\varepsilon} \quad \text{(Gauss's Law)}$$
$$E = \frac{\lambda}{2\pi\varepsilon r} \quad \text{(Infinite line charge)}$$
$$E = \frac{\sigma}{2\varepsilon} \quad \text{(Infinite thin sheet)}$$
$$E = \frac{\sigma}{\varepsilon} \quad \text{(Thick sheet / Cond. surface)}$$

- λ: Linear charge density ($\mathrm{C\,m^{-1}}$)
- σ: Surface charge density ($\mathrm{C\,m^{-2}}$)
- ρ: Volume charge density ($\mathrm{C\,m^{-3}}$)
- Φ: Electric flux (V m)

6.3 Potential & Work

$$V = \frac{1}{4\pi\varepsilon}\frac{Q}{r} \quad \text{(Potential by Q)}$$
$$W = Vq \quad \text{(Potential Energy)}$$
$$E = -\frac{\Delta V}{d} = -\frac{dV}{dx} \quad \text{(Potential grad.)}$$

- V: Electric potential (V or $\mathrm{J\,C^{-1}}$)
- W: Work done moving charge q (J)
- d, x: Distance (m)

6.4 Capacitors

$$Q = CV$$
$$C = 4\pi\varepsilon R \quad \text{(Isolated sphere)}$$
$$C = \frac{\varepsilon A}{d} \quad \text{(Parallel plate capacitor)}$$
$$W = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C} \quad \text{(Energy)}$$
$$C_{eq} = C_1 + C_2 \quad \text{(Parallel)}$$
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \quad \text{(Series)}$$

- C: Capacitance (F)
- A: Overlapping area of plates ($\mathrm{m^2}$)
- d: Separation between plates (m)

7 MAGNETIC FIELDS

7.1 Magnetic Force & Flux

$$\Phi = BA$$
$$F = BIL \quad \text{(Conductor in B-field)}$$
$$F = Bqv \quad \text{(Moving charge)}$$
$$\tau = BIAN\cos\theta$$

- Φ: Magnetic flux (Wb)
- B: Magnetic flux density / Field (T)
- *Direction:* Fleming's Left-Hand Rule for Force
- τ: Torque on a coil (N m)
- θ (in torque): Angle between coil *plane* and B
- N: Number of turns

7.2 Magnetic Field Sources

$$\Delta B = \frac{\mu_0}{4\pi}\frac{I\Delta L\sin\theta}{r^2} \quad \text{(Biot-Savart Law)}$$
$$B = \frac{\mu_0 I}{2\pi r} \quad \text{(Infinite straight wire)}$$
$$B = \frac{\mu_0 I}{2r} \quad \text{(Center of circular loop)}$$
$$B = \mu_0 nI = \mu_0\left(\frac{N}{L}\right)I \quad \text{(Solenoid)}$$

- μ_0: Permeability of free space ($4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$)
- r: Radial distance / radius (m)
- n: Number of turns per unit length
- *Direction:* Right-Hand Grip Rule

7.3 Electromagnetic Induction & Hall Effect

$$E = -\frac{d\Phi}{dt} \quad \text{(Faraday's \& Lenz's Law)}$$
$$E = Blv \quad \text{(EMF - straight rod)}$$
$$E = \frac{1}{2}BL^2\omega \quad \text{(Rotating rod)}$$
$$V_H = \frac{BI}{nte} \quad \text{(Hall Voltage)}$$

- E: Induced EMF (V)
- v: Velocity ($\mathrm{m\,s^{-1}}$), ω: Angular velocity ($\mathrm{rad\,s^{-1}}$)
- V_H: Hall voltage (V)
- t: Thickness of the conductor (m)
- n: Charge carrier density ($\mathrm{m^{-3}}$)

8 CURRENT ELECTRICITY

8.1 Current & Resistance

$$I = \frac{Q}{t}$$
$$J = \frac{I}{A} = nev_d \quad \text{(Current density)}$$
$$v_d = \frac{I}{nAe}$$
$$R = \frac{\rho L}{A}$$
$$R_\theta = R_0(1 + \alpha\Delta\theta)$$

- I: Current (A), Q: Charge (C)
- v_d: Drift velocity ($\mathrm{m\,s^{-1}}$)
- e: Elementary charge (1.6×10^{-19} C)
- R: Resistance (Ω)
- ρ: Resistivity (Ω m)
- α: Temp. coefficient of resistance ($^\circ\mathrm{C^{-1}}$)

8.2 Circuits & Power

$$R_{eq} = R_1 + R_2 \quad \text{(Series)}$$
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad \text{(Parallel)}$$
$$P = VI = I^2R = \frac{V^2}{R} \quad \text{(Power)}$$
$$\frac{V_1}{V_2} = \frac{R_1}{R_2} \quad \text{(Potential Divider - Series)}$$
$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \quad \text{(Current Divider - Parallel)}$$

- P: Power dissipated (W)

8.3 Kirchhoff's Laws & Batteries

$$\sum I_{in} = \sum I_{out} \quad \text{(KCL)}$$
$$\sum E = \sum IR \quad \text{(KVL)}$$
$$V = E - Ir \quad \text{(Disch. battery)}$$
$$V = E + Ir \quad \text{(Charging battery)}$$

- E: Electromotive force (EMF) of battery (V)
- V: Terminal potential difference (V)
- r: Internal resistance (Ω)

8.4 Measuring Instruments

Potentiometer:

$$E \propto l \implies \frac{E_1}{E_2} = \frac{l_1}{l_2}$$
$$r = R\left(\frac{l_1}{l_2} - 1\right) \quad \text{(Finding r)}$$

Meter Bridge:

$$\frac{R_1}{R_2} = \frac{l_1}{l_2} = \frac{l}{100 - l} \quad \text{(At Balance)}$$

- l: Balance length (cm or m)

9 ELECTRONICS

9.1 Diodes & Transistors (BJT)

- **Potential Barrier:** Si $\approx$ 0.7V, Ge $\approx$ 0.3V
- **Diodes:** Forward bias (conducts), Reverse bias (blocks). Zener diodes operate in reverse breakdown for voltage regulation.

$$I_E = I_C + I_B$$
$$\beta = \frac{I_C}{I_B} \quad \text{(Common-emitter gain)}$$
$$\alpha = \frac{I_C}{I_E} \quad \text{(Common-base gain)}$$

BJT Operating Regions:

- **Cut-off:** $V_{BE} < 0.7$V, $I_B = 0, I_C = 0$ (Switch OFF)
- **Active:** $V_{BE} \approx 0.7$V, $V_{CE} > 0$ (Amplifier)
- **Saturation:** I_C is max, $V_{CE} \approx 0$V (Switch ON)

9.2 Operational Amplifiers (Op-Amps)

$$V_{out} = A_0(V_+ - V_-) \quad \text{(Open Loop)}$$
$$\text{If } V_+ > V_-, \quad V_{out} = +V_S \quad \text{(+ Sat.)}$$
$$\text{If } V_+ < V_-, \quad V_{out} = -V_S \quad \text{(- Sat.)}$$

Closed Loop (Negative Feedback):

$$V_{out} = \left(-\frac{R_f}{R_{in}}\right)V_{in} \quad \text{(Inverting)}$$
$$V_{out} = \left(1 + \frac{R_f}{R_{in}}\right)V_{in} \quad \text{(Non-invt.)}$$

9.3 Digital Electronics

Logic Gates:

- AND: $F = A \cdot B$ OR: $F = A + B$
- NOT: $F = \overline{A}$ NAND: $F = \overline{A \cdot B}$
- NOR: $F = \overline{A + B}$

Boolean Identities:

$$\overline{\overline{A}} = A$$
$$A \cdot \overline{A} = 0, \quad A + \overline{A} = 1$$
$$A + 1 = 1, \quad A \cdot 1 = A$$
$$A + 0 = A, \quad A \cdot 0 = 0$$
$$A \cdot (B + C) = A \cdot B + A \cdot C$$

De Morgan's Laws:

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$
$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

SR Flip-Flop (Active High):

- $S = 0, R = 0 \implies Q_{next} = Q_{prev}$ (Hold)
- $S = 0, R = 1 \implies Q = 0$ (Reset)
- $S = 1, R = 0 \implies Q = 1$ (Set)
- $S = 1, R = 1 \implies$ Invalid / Forbidden

10 PROPERTIES OF MATTER

10.1 Elasticity

$$\text{Stress } (\sigma) = \frac{F}{A}$$
$$\text{Strain } (\varepsilon) = \frac{e}{L}$$
$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{FL}{Ae}$$
$$F = YA\alpha\Delta\theta \quad \text{(Thermal Force)}$$
$$W = \frac{1}{2}Fe \quad \text{(Strain Energy)}$$
$$U_v = \frac{1}{2}\sigma\varepsilon \quad \text{(E per unit vol.)}$$

- Y: Young's Modulus (Pa)
- e: Extension (m), L: Original length (m)
- α: Linear expansivity ($^\circ\mathrm{C^{-1}}$)

10.2 Viscosity

$$F = \eta A\frac{\Delta v}{\Delta d} \quad \text{(Newton's Law)}$$
$$\frac{V}{t} = \frac{\pi r^4\Delta P}{8\eta L} \quad \text{(Poiseuille's Equation)}$$
$$F = 6\pi\eta rv \quad \text{(Stokes' Law)}$$

- η: Coefficient of viscosity (Pa s)
- $\Delta v/\Delta d$: Velocity gradient ($\mathrm{s^{-1}}$)
- V/t: Volume flow rate ($\mathrm{m^3\,s^{-1}}$)
- r: Radius of tube/sphere (m)
- ΔP: Pressure difference (Pa)

10.3 Surface Tension

$$\Delta P = \frac{2T}{r} \quad \text{(Excess pressure)}$$
$$\Delta P = \frac{4T}{r} \quad \text{(Soap bubble)}$$
$$h\rho g = \frac{2T\cos\theta}{r} \quad \text{(Capillary rise)}$$
$$W = T\Delta A \quad \text{(Surface Energy)}$$

- T: Surface Tension ($\mathrm{N\,m^{-1}}$)
- r: Radius of drop / capillary tube (m)
- h: Height of liquid column (m)
- θ: Contact angle
- ΔA: Change in surface area

11 MATTER & RADIATION

11.1 Thermal Radiation

$$I = \sigma eT^4 \quad \text{(Intensity)}$$
$$P = \sigma eAT^4 \quad \text{(Power radiated)}$$
$$P_{net} = \sigma eA(T^4 - T_s^4) \quad \text{(Net power)}$$
$$E = Pt = \sigma eAT^4t \quad \text{(Tot. Energy)}$$
$$\lambda_{\max}T = c \quad \text{(Wien's Disp. Law)}$$

- I: Intensity ($\mathrm{W\,m^{-2}}$)
- P: Power (W), E: Energy (J)
- σ: Stefan-Boltzmann const (5.67×10^{-8})
- e: Emissivity ($e = 1$ for black body)
- T_s: Surrounding temperature (K)
- c: Wien's constant ($\approx 2.898 \times 10^{-3}$ m K)

11.2 Quantum Mechanics & Photo-electric Effect

$$E = hf = \frac{hc}{\lambda} \quad \text{(Photon Energy)}$$
$$\phi = hf_0 = \frac{hc}{\lambda_0} \quad \text{(Work Function)}$$
$$hf = \phi + K_{\max} \quad \text{(Einstein's Eq.)}$$
$$K_{\max} = \frac{1}{2}mv^2 = eV_s$$
$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad \text{(de Broglie)}$$

- h: Planck's constant (6.63×10^{-34} J s)
- f, f_0: Frequency / Threshold Freq. (Hz)
- λ, λ_0: Wavelength / Threshold (m)
- $K_{\max}$: Max Kinetic Energy (J or eV)
- V_s: Stopping potential (V)
- p: Momentum ($\mathrm{kg\,m\,s^{-1}}$)

11.3 X-Rays

$$eV = \frac{hc}{\lambda_{\min}} \quad \text{(Max X-Ray photon E.)}$$
$$\lambda_{\min} = \frac{hc}{eV} \quad \text{(Minimum wavelength)}$$

- V: Accelerating voltage (V)

11.4 Radioactivity & Nucleus

$$A = \frac{dN}{dt} = \lambda N \quad \text{(Activity)}$$
$$N = N_0e^{-\lambda t} \quad \text{(For Nuclei)}$$
$$A = A_0e^{-\lambda t} \quad \text{(For Activity)}$$
$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda} \quad \text{(Half-life)}$$
$$E = \Delta mc^2 \quad \text{(Mass-Energy Equiv.)}$$

- A: Activity (Bq or decays per second)
- λ: Decay constant ($\mathrm{s^{-1}}$)
- N: Number of undecayed nuclei
- $T_{1/2}$: Half-life (s)
- E: Energy released in fission/fusion (J or MeV)
- c: Speed of light ($3 \times 10^8\,\mathrm{m\,s^{-1}}$)